Name: _____.

Student number: _____.

Instructions

- Turn off all the electronic devices.
- This is a closed book exam.
- You are allowed a notesheet. You can use any fact in the notesheet, but you must reference which fact you are using.
- You must justify your answers.
- You will have 60 minutes to complete the exam.
- The exam will be similar in content, length and difficulty to the first three questions below. The remaining questions should be used for extra practice.
- Good luck!

Question:	1	2	3	4	5	6	7	Total
Points:	10	10	10	10	10	10	10	70
Score:								

1. (10 points) Consider an equivalence relation $\sim$ on a topological space X. Prove that the following are equivalent

   (1) The quotient map $\pi: X \to X/\sim$ is open.

   (2) For every open subset $U \subseteq X$, the union of all equivalence classes that meet U is open in X.

Before we start, for any $U \subseteq X$, let $P_U = \{ a \in X \mid [a] \cap U \neq \emptyset \}$

Then the union of all eq. classes that meet U is the set

$$S_U = \bigcup_{a \in P_U} [a]$$

Claim: $S_U = \pi^{-1}(\pi(U))$

[$\subseteq$] Given $v \in S_U$, we have $v \in [a]$ for some $a \in P_U$. Hence $\pi(v) = \pi(a)$. But since $[a] \cap U \neq \emptyset$, there is some $u \in [a] \cap U$

Since $u \in [a]$, we have $\pi(u) = \pi(a)$. Therefore, since $u \in U$,

$$\pi(v) = \pi(a) = \pi(u) \in \pi(U)$$

Thus $\pi(v) \in \pi(U)$, which means that $v \in \pi^{-1}(\pi(U))$

[$\supseteq$] Given $x \in \pi^{-1}(\pi(U))$ we have $\pi(x) \in \pi(U)$. Hence, there is some $u \in U$ such that $\pi(x) = \pi(u)$, or in other words $x \in [u]$

Since $u \in U$ we have $u \in [u] \cap U$, so $[u] \cap U \neq \emptyset$ and $u \in P_U$

Therefore $x \in [u] \subseteq \bigcup_{a \in P_U} [a] = S_U$ $\square$ claim

Now we prove the equivalence (1) $\Leftrightarrow$ (2)

1. (10 points) Consider an equivalence relation $\sim$ on a topological space X. Prove that the following are equivalent

   (1) The quotient map $\pi: X \to X/\sim$ is open.

   (2) For every open subset $U \subseteq X$, the union of all equivalence classes that meet U is open in X.

(1) $\Rightarrow$ (2) Assume $U \subseteq X$ is open Since π is open we have

$\pi(U)$ open in $X/_\sim$. Since π is continuous

$S_U = \pi^{-1}(\pi(U))$ is open in X.

(2) $\Rightarrow$ (1) Assume $V \subseteq X$ is open. We want To show

$\pi(V)$ is open in $X/_\sim$. By defn of The quotient TOP.

That means That $\pi^{-1}(\pi(V))$ is open in X

But we are assuming That $S_V = \pi^{-1}(\pi(V))$ is open

(claim)

whenever V is open. Hence π is an open map.

2. (10 points) Let X be a **connected** topological space. Let f_1 and f_2 be continuous maps $f_1, f_2 : X \to \mathbb{R}$. Assume that, for each $x \in X$, we have $f_1(x) \neq f_2(x)$.
Prove that either $f_1(x) < f_2(x)$ for all $x \in X$, or $f_1(x) > f_2(x)$ for all $x \in X$.

Let $V_1 = \{x \in X \mid f_1(x) < f_2(x)\}$ and $V_2 = \{x \in X \mid f_1(x) > f_2(x)\}$

Of course, $V_1 \cap V_2 = \emptyset$. Also, since $f_1(x) \neq f_2(x)$ $\forall x \in X$, we have $X = V_1 \cup V_2$.

Furthermore, since $d = f_1 - f_2 : X \longrightarrow \mathbb{R}$ is continuous (Problem 4 below), we know that $V_1 = d^{-1}(-\infty, 0)$ and $V_2 = d^{-1}(0, \infty)$ are both open in X because $(-\infty, 0)$ and $(0, \infty)$ are open in $\mathbb{R}$.

Hence $X = V_1 \cup V_2$ with $V_1 \cap V_2 = \emptyset$ and V_1, V_2 open.

Since X is connected, either $V_1 = \emptyset$ or $V_2 = \emptyset$.

In case $V_1 = \emptyset$ we have $f_1(x) > f_2(x)$ $\forall x \in X$.

In case $V_2 = \emptyset$ we have $f_1(x) < f_2(x)$ $\forall x \in X$.

□

3. (10 points) Let X be a **compact** topological space. Let $g: X \to \mathbb{R}$ and $f_n: X \to \mathbb{R}$ be continuous maps, where $n \in \mathbb{N}$.

   Assume that, for each $x \in X$, we have $f_n(x) < f_{n+1}(x)$ and $\lim_{n\to\infty} f_n(x) = f(x)$.

   Prove that $\lim_{n\to\infty} \sup\{|f(x) - f_n(x)| : x \in X\} = 0$.

Take $\varepsilon > 0$. We want to show that there exists some $N > 0$ such that, if $n \geq N$ then $\sup\{|f(x) - f_n(x)| : x \in X\} < \varepsilon$

Or, equivalently, such that, if $n \geq N$, then $|f(x) - f_n(x)| < \varepsilon \quad \forall x \in X$

Fix $x \in X$. Note some facts:

1) Since $\lim_{n\to\infty} f_n(x) = f(x)$, for this given $\varepsilon > 0$, there is some $N_x \in \mathbb{N}$ such that if $n \geq N_x$ then $|f(x) - f_n(x)| < \frac{\varepsilon}{3}$ (for this x)

2) Since f_{N_x} and f are continuous maps, there is an open nbhd U_x of x such that, for any $y \in U_x$, we have

$$|f_{N_x}(y) - f_{N_x}(x)| < \frac{\varepsilon}{3} \quad \text{and} \quad |f(y) - f(x)| < \frac{\varepsilon}{3}$$

(In principle, one nbhd works for f_{N_x} and another one works for f, but we can assume they are the same by taking their intersection)

Hence, for any $y \in U_x$, by the Δ-inequality (twice) we have

$$|f_{N_x}(y) - f(y)| \leq |f_{N_x}(y) - f_{N_x}(x)| + |f_{N_x}(x) - f(x)| + |f(x) - f(y)|$$
$$< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon$$

3) Since $f_n(y) < f_{n+1}(y)$ $\forall n$ and $f_n(y) \to f(y)$, we have $f_n(y) \leq f(y)$ $\forall n$

Hence, $\forall\, n \geq N_x$, we have $f_{N_x}(y) \leq f_n(y) \leq f(y)$

Thus for any $y \in U_x$ we have, for any $n \geq N_x$,

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3. (10 points) Let X be a **compact** topological space. Let $g: X \to \mathbb{R}$ and $f_n: X \to \mathbb{R}$ be continuous maps, where $n \in \mathbb{N}$.

   Assume that, for each $x \in X$, we have $f_n(x) < f_{n+1}(x)$ and $\lim_{n\to\infty} f_n(x) = f(x)$.

   Prove that $\lim_{n\to\infty} \sup\{|f(x) - f_n(x)| : x \in X\} = 0$.

$|f(y) - f_n(y)| \leq |f(y) - f_{N_x}(y)| < \varepsilon$

To summarize, for each $x \in X$ there exist an open $x \in U_x \subseteq X$ and a positive integer N_x such that

$$|f(y) - f_n(y)| < \varepsilon \quad \forall\, y \in U_x \quad \forall\, n \geq N_x.$$

Since $\{U_x \mid x \in X\}$ forms an open cover of X, which is compact, we can pass to a finite subcover. Thus

$X = U_{x_1} \cup U_{x_2} \cup \cdots \cup U_{x_m}$ for some $x_1, \ldots, x_m \in X$.

Let $N = \max\{N_{x_1}, N_{x_2}, \ldots, N_{x_m}\}$

For an arbitrary $y \in X$, we have $y \in U_{x_j}$ for some j, and since $N \geq N_{x_j}$, we know that for any $n \geq N (\geq N_{x_j})$ we have

$$|f(y) - f_n(y)| < \varepsilon \quad \text{(because } y \in U_{x_j} \text{ and } n \geq N_{x_j}\text{)}$$

This works for any $y \in X$, so

$$|f(y) - f_n(y)| < \varepsilon \quad \forall\, n \geq N, \forall y \in X.$$

Hence $\sup\{|f(y) - f_n(y)| : y \in X\} \leq \varepsilon \quad \forall\, n \geq N.$

Thus $\lim_{n\to\infty} \sup\{|f(y) - f_n(y)| : y \in X\} = 0.$ □

Extra Practice

4. (10 points) Let X be a topological space. Let $f\colon X \to \mathbb{R}$ and $g\colon X \to \mathbb{R}$ be continuous maps. Prove the following

   (a) The map $s\colon X \to \mathbb{R}$ given by $s(x) = f(x) + g(x)$ is continuous.

   (b) The map $p\colon X \to \mathbb{R}$ given by $p(x) = f(x)g(x)$ is continuous.

   (c) The map $d\colon X \to \mathbb{R}$ given by $d(x) = f(x) - g(x)$ is continuous.

   Hint: prove first that the maps $\mathbb{R}^2 \to \mathbb{R}$ where $(x, y) \mapsto x + y$ and $(x, y) \mapsto xy$ are continuous.

5. (10 points) Let $\{V_i \mid i \in \Lambda\}$ be a collection of connected subspaces of topological space X.
   Assume that there exists some $j \in \Lambda$ such that $V_j \cap V_i \neq \varnothing$ for all $i \in \Lambda$.
   Prove that $\bigcup_{i\in\Lambda} V_i$ is connected.

6. (10 points) Let $\omega \notin \mathbb{R}$ and define $X = \mathbb{R} \cup \{\omega\}$. For each $x \in X$ and $r > 0$, define

$$A(x, r) = \begin{cases} \{y \in \mathbb{R} \mid |y - x| < r\}, & \text{if } x \in \mathbb{R} \\ \{y \in \mathbb{R} \mid |y| > r\} \cup \{\omega\}, & \text{if } x = \omega. \end{cases}$$

   We showed in the Practice Midterm that $\mathcal{A} = \{A(x, r) \mid x \in X,\ r > 0\}$ is a basis for a topology on X.
   Prove that X is compact.

7. (10 points) Let X be a topological space. Decide if the following statements are true or false.

   (a) Any finite union of compact subspaces of X is compact.

   (b) Any finite intersection of compact subspaces of X is compact.